

Joshua Urban Davis

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I am a research scientist and software engineer with 6 years of experience currently working on machine learning and computer vision applications to AR/VR at Meta. I previously worked at Adobe Research in the GILL lab and held internships at Microsoft Research and Autodesk during my PhD studies in computer science at Dartmouth. Prior to graduate school I worked at several software startups including the DALI Lab, Bregmann Media Labs, and Hex Plexus between 2013 - 2017.

Languages: Python, C++, JavaScript, C#, MATLAB

Frameworks & Libraries: PyTorch, NumPy, OpenCV, MediaPipe, OpenGL, TensorFlow, scikit-learn, Pandas

Developer Tools: Git, Docker, Unity, Jira, CI/CD pipelines

Specialties: lighting estimation, semantic scene understanding, 3D reconstruction, SLAM, on-device inference, generative AI, AR/VR



META | REALITY LAB

Research Scientist / Software Engineer

Menlo Park, CA

Current role

- Designed, implemented, and optimized computer vision and machine learning pipelines for **real-time lighting estimation** and **photorealistic virtual object rendering** in AR/VR/MR.
- Engineered **color perception algorithms** including metric design, KPI evaluation pipelines, and integration into camera + AR software stacks.
- Developed **3D surface reconstruction** and **semantic scene understanding** algorithms using SLAM, SIFT, and pix2pix for novel lighting estimation features
- Built and maintained **large-scale dataset generation and preprocessing pipelines** in Python, PyTorch, and OpenCV to accelerate model development
- Collaborated with cross-functional engineering teams to **deploy research prototypes into production-grade AR/VR software**.



ADOBE

Research Assistant

San Jose, CA

2021 - 2022

- Developed an interactive, **real-time media augmentation system** for video conferencing with body pose recognition, gesture detection, and speech processing.
- Prototyped end-to-end features in Python and JavaScript, leveraging **MediaPipe** and **OpenCV**, with performance optimizations for on-device inference.
- Refactored and optimized **object detection** and **3D human pose estimation** models for deployment on mobile and desktop
- Designed and implemented **automated data capture and analysis pipelines** for large-scale user studies, enabling rapid iteration and actionable product insights.



MICROSOFT RESEARCH

Research Internship

Redmond, WA

2020

- Built an **interactive mask system** using a smartphone to display real-time video of the wearer's mouth and nose on the mask surface.
- Implemented real-time mapping algorithms for **distortion-free visualization** and optimized for **low-latency on-device inference**
- Collaborated with hardware and software engineers to transition the prototype into a **commercial-ready product**.



AUTODESK

Research Internship

Toronto, ON

2019

- Developed **generative AI tools** for authoring 3D objects in VR using GANs, Transformers, and Autoencoders
- Implemented interactive generative models in PyTorch with **Unity** integration for immersive design workflows.
- Optimized model performance for **real-time VR interaction** and contributed to patented VR interaction techniques.

EDUCATION

DARTMOUTH COLLEGE

Doctorate of Computer Sciences

Dissertation Topic: Multimodal Applications of Generative AI and Computer Vision to AR/VR and 3D Geometry

Hanover, NH

PATENTS & ADDITIONAL SKILLS

- (Calliope) A System for Supporting Human-AI Collaboration in Virtual Environments. Josh Urban Davis, Fraser Anderson, George Fitzmaurice. (Pending, No. 076/0301)
- (TangibleCircuits) A System for Converting Circuit Diagrams to Tangible and Audio Interfaces. Josh Urban Davis, Xing-Dong Yang (No. 61/030,441)
- (Circuit Style) A System for Peripherally Reinforcing Best Practices in Hardware Computing. Josh Urban Davis, Jun Gong, Xing-Dong Yang (No. 62/916,977)